Technical Appendix

Synthetic Cohort Modeling of Seminary Formation Attrition

Dataset Structure, Generation Logic, Dashboard Configuration, and Validation Documentation

Technical Overview

This appendix documents the technical structure of the synthetic cohort dataset, dashboard configuration, variable architecture, generation logic, and validation procedures used within the Seminary Formation Attrition project.

The analytical environment was constructed using:

- Microsoft Excel
- conditional spreadsheet formulas
- randomized bounded generation logic
- pivot-table review structures
- Power BI dashboard reporting
- DAX-based label formatting

The project was developed as a synthetic institutional-analysis and dashboard-reporting environment rather than a predictive statistical model.

Dataset Architecture

The final dataset contained 300 synthetic observations representing simulated seminary formation trajectories.

The dataset was organized into five primary variable categories:

- background variables
- behavioral indicators
- institutional process indicators
- outcome variables
- trajectory typologies

The workbook structure included:

- generator logic worksheets
- static dataset worksheets
- pivot-table review sheets
- variable codebook documentation
- dashboard staging tables

Synthetic observations were initially generated using live spreadsheet formulas. Formula outputs were later copied into a static dataset worksheet prior to import into Power BI in order to stabilize dashboard reporting outputs and prevent recalculation drift from randomized formulas.

Figure 1 documents the variable codebook structure and analytical field organization.

Figure 1) Variable Codebook Structure

Synthetic Cohort Modeling of Seminary Formation Attrition

Variable	Type	Coding / Definition	Analytical Role
Behavioral and Dispositional Indicators			
conscientiousness_score	Likert (1–5)	1=very low, 3=moderate, 5=very high; self-discipline/reliability	Behavioral stability proxy
internal_locus_of_control	Likert (1–5)	1=externalized, 3=moderate, 5=internalized; perceived agency over	Agency orientation proxy
introspection_level	Likert (1–5)	1=low reflective interior scrutiny, 5=high introspective	Reflective disposition proxy
compliance_score	Likert (1–5)	Behavioral conformity under institutional evaluation; 1=low, 5=high	Institutional adaptability proxy
willingness_to_self_edit	Likert (1–5)	Willingness to modify behavior or presentation in response to	Adaptive flexibility proxy
conscience_conflict_events	Integer (count proxy)	Synthetic proxy for accumulated conscience or institutional-tension	Internal conflict accumulation indicator
initial_evaluation_score	Likert (1–5)	Synthetic proxy for strength of initial institutional evaluation; 1=low,	Initial institutional assessment proxy
Institutional Process Indicators			
year1_advancement	Binary (0/1)	0=no, 1=yes; advanced past year-1 checkpoint	Early-stage progression indicator
year2_advancement	Binary (0/1)	0=no, 1=yes; advanced past year-2 checkpoint	Mid-stage progression indicator
leadership_flag	Binary (0/1)	0=no, 1=yes; assigned/recognized leadership responsibility	Institutional recognition indicator
Outcome Variables			
ordination_outcome	Binary (0/1)	0=not ordained, 1=ordained	Terminal institutional outcome
year_of_exit	Integer (years since entry)	Years after entry when exit occurred; blank/NA if ordained	Attrition timing indicator
exit_reason_category	Integer (1–5)	The following variables were operationalized to support synthetic cohort construction, institutional segmentation analysis, and descriptive dashboard reporting; blank/NA if ordained:	Attrition segmentation variable
Exit Reason Categories			
Code	**Definition**		
1	Loss of faith		
2	Relational or sexual factors		
3	Psychological strain		
4	Institutional mismatch		
5	Academic difficulty		
Trajectory Typology			
profile_cluster	Category (typology)	Synthetic formation-trajectory typology (partly outcome-linked): early_compliant_survivor | institutional_careerist | early_exit_discerner | late_integrated_attriter	Primary segmentation framework
Methodological Note			
Behavioral, institutional, and typological variables within this framework function as synthetic analytical proxies rather than validated psychometric, diagnostic, or clinical measures. Variables were operationalized to support descriptive segmentation analysis and institutional-process modeling within a rule-based synthetic cohort environment.			

Background Variables

Variable	Type	Description
id	Integer	Unique row identifier
age_at_entry	Integer	Age at seminary entry
entry_cohort_year	Integer	Simulated cohort entry year

prior_career_years	Integer	Prior work or career experience
diocesan_or_order	Category	Institutional affiliation type
region	Category	Regional assignment category

Behavioral Indicators

Variable	Type	Description
conscientiousness_score	Likert (1–5)	Behavioral stability proxy
internal_locus_of_control	Likert (1–5)	Agency orientation proxy
introspection_level	Likert (1–5)	Reflective disposition proxy
compliance_score	Likert (1–5)	Institutional adaptability proxy
willingness_to_self_edit	Likert (1–5)	Adaptive flexibility proxy
conscience_conflict_events	Integer	Institutional tension/conflict proxy
initial_evaluation_score	Likert (1–5)	Initial institutional assessment proxy

Institutional Process Indicators

Variable	Type	Description
year1_advancement	Binary (0/1)	Early-stage progression indicator
year2_advancement	Binary (0/1)	Mid-stage progression indicator
leadership_flag	Binary (0/1)	Institutional recognition indicator

Outcome Variables

Variable	Type	Description
ordination_outcome	Binary (0/1)	Terminal institutional outcome
year_of_exit	Integer	Attrition timing indicator
exit_reason_category	Integer (1–5)	Attrition segmentation category

Trajectory Typology Variable

Variable	Type	Description
profile_cluster	Category	Primary segmentation typology

Trajectory typologies included:

- Early_compliant_survivor
- Institutional_careerist
- Early_exit_discerner
- late_integrated_attriter

Sample Generation Logic

Synthetic observations were generated using conditional spreadsheet formulas and typology-conditioned scoring structures.

Examples of generation logic included:

- age ranges conditioned by trajectory distribution
- advancement checkpoints conditioned by prior outcomes
- ordination outcomes gated through compliance, introspection, and conflict variables
- exit timing conditioned by attrition profile structure

Representative generation logic included:

- randomized bounded scoring ranges using RANDBETWEEN()
- conditional assignment using IF() statements
- profile segmentation using SWITCH() and IFS() logic
- outcome gating using nested logical conditions

Generation logic was revised iteratively during development to improve internal consistency between advancement indicators, terminal outcomes, and dashboard segmentation behavior.

Figure 2 documents the static dataset structure and generated variable outputs.

Figure 2) Static Dataset Structure

id	age_at_entry	entry_cohort_yea	prior_career_yea	diocesan_or_orc	region	conscientiousne	internal_locus_o	introspection_le	compliance_sco	willingness_to_s	conscience_con	initial_evaluation	year1_advancen	year2_advancen	leadership_flag	year_of_exit	ordination_outco	exit_reason_cate	profile_cluster
1	42	2004	20	religious	Midwest	5	2	1	5	5	1	3	1	1	0		1		institutional_careerist
2	34	2012	12	religious	South	4	3	2	5	5	0	3	1	1	0		1		institutional_careerist
3	24	2022	2	diocesan	West	3	1	2	3	2	1	3	0	0	0	2	0	1	early_exit_discerner
4	35	2011	13	religious	West	5	2	2	4	5	0	4	1	1	1		1		institutional_careerist
5	29	2017	7	diocesan	Midwest	4	5	4	1	1	3	4	1	1	1	3	0	4	late_integrated_attriter
6	27	2019	5	diocesan	Midwest	4	3	3	4	4	1	4	1	1	1		1		early_compliant_survivor
7	27	2019	5	diocesan	West	5	3	3	3	4	1	4	1	1	1		1		early_compliant_survivor
8	27	2019	5	diocesan	Northeast	5	3	3	3	5	1	4	1	1	1		1		early_compliant_survivor
9	31	2015	9	diocesan	West	3	5	5	1	2	4	4	1	1	1	6	0	3	late_integrated_attriter
10	24	2022	2	diocesan	South	5	3	3	4	4	1	4	1	1	1		1		early_compliant_survivor
11	26	2020	4	diocesan	West	5	3	3	4	5	0	4	1	1	1		1		early_compliant_survivor
12	41	2005	19	religious	Midwest	4	2	2	5	4	0	3	1	1	0		1		institutional_careerist
13	34	2012	12	religious	Midwest	4	2	2	4	4	0	3	1	1	0		1		institutional_careerist
14	24	2022	2	diocesan	West	2	2	2	3	3	1	2	0	0	0	2	0	5	early_exit_discerner
15	35	2011	13	religious	West	4	2	1	4	4	1	3	1	1	0		1		institutional_careerist
16	26	2020	4	diocesan	Northeast	3	2	2	2	2	2	3	0	0	0	2	0	1	early_exit_discerner
17	28	2018	6	diocesan	Midwest	5	4	4	3	5	2	5	1	1	1	3	0	3	early_compliant_survivor
18	34	2012	12	diocesan	Midwest	4	5	5	1	2	4	5	1	1	1	4	0	4	late_integrated_attriter
19	38	2008	16	religious	Northeast	5	2	2	4	5	0	4	1	1	1		1		institutional_careerist
20	23	2023	1	diocesan	South	4	4	3	4	5	0	4	1	1	1		1		early_compliant_survivor
21	25	2021	3	diocesan	South	4	3	4	3	4	2	4	1	1	1	3	0	3	early_compliant_survivor
22	38	2008	16	diocesan	West	3	4	4	2	1	5	4	1	1	1	6	0	3	late_integrated_attriter
23	29	2017	7	diocesan	South	5	3	3	4	5	0	4	1	1	1		1		early_compliant_survivor
24	26	2020	4	diocesan	Midwest	4	3	3	4	4	1	4	1	1	1		1		early_compliant_survivor
25	24	2022	2	diocesan	West	5	4	4	4	5	2	5	1	0	1	2	0	3	early_compliant_survivor
26	25	2021	3	diocesan	South	5	4	3	4	4	1	4	1	1	1		1		early_compliant_survivor
27	45	2001	23	religious	Midwest	5	2	1	5	5	1	3	1	1	0		1		institutional_careerist
28	40	2006	18	religious	Midwest	5	2	1	4	5	1	3	1	1	0		1		institutional_careerist
29	29	2017	7	diocesan	South	4	3	4	4	5	2	4	1	0	1	2	0	3	early_compliant_survivor
30	41	2005	19	religious	Northeast	4	3	1	5	4	0	3	1	1	0		1		institutional_careerist
31	28	2018	6	diocesan	South	4	3	3	4	5	1	4	1	1	1		1		early_compliant_survivor
32	35	2011	13	religious	West	4	2	1	4	5	1	3	1	1	0		1		institutional_careerist
33	34	2012	12	religious	Midwest	4	3	1	4	4	0	3	1	1	0		1		institutional_careerist
34	25	2021	3	diocesan	Northeast	2	1	3	2	3	2	3	0	0	0	1	0	5	early_exit_discerner
35	43	2003	21	religious	Northeast	5	2	1	4	5	1	3	1	1	0		1		institutional_careerist
36	40	2006	18	diocesan	West	4	5	4	2	1	5	4	1	1	1	4	0	3	late_integrated_attriter
37	33	2013	11	religious	West	4	3	2	4	4	0	3	1	1	0		1		institutional_careerist
38	29	2017	7	diocesan	Midwest	3	4	5	1	1	4	4	1	1	1	4	0	4	late_integrated_attriter
39	28	2018	6	diocesan	Northeast	4	4	3	4	4	0	4	1	1	1		1		early_compliant_survivor
40	24	2022	2	diocesan	West	3	2	2	3	3	2	3	0	0	0	2	0	1	early_exit_discerner
41	22	2024	0	diocesan	West	2	2	2	3	3	1	2	0	0	0	1	0	1	early_exit_discerner
42	27	2019	5	diocesan	West	5	4	3	3	5	0	4	1	1	1		1		early_compliant_survivor
43	25	2021	3	diocesan	South	4	4	4	4	4	2	4	1	0	1	2	0	3	early_compliant_survivor
44	27	2019	5	diocesan	Northeast	5	3	4	4	4	2	5	1	0	1	2	0	3	early_compliant_survivor
45	29	2017	7	diocesan	Northeast	4	4	3	3	5	0	4	1	1	1		1		early_compliant_survivor
46	25	2021	3	diocesan	Midwest	3	2	2	2	2	2	3	0	0	0	1	0	1	early_exit_discerner
47	29	2017	7	diocesan	Midwest	4	4	4	4	5	1	4	1	0	1	2	0	3	early_compliant_survivor
48	29	2017	7	diocesan	West	5	4	3	4	5	0	4	1	1	1		1		early_compliant_survivor
49	38	2008	16	diocesan	Northeast	4	5	5	2	2	5	5	1	1	1	5	0	3	late_integrated_attriter
50	26	2020	4	diocesan	Northeast	2	2	3	3	3	1	3	0	0	0	1	0	1	early_exit_discerner
51	27	2019	5	diocesan	West	5	4	3	4	4	0	4	1	1	1		1		early_compliant_survivor
52	34	2012	12	religious	Northeast	5	3	2	4	4	1	4	1	1	1		1		institutional_careerist
53	24	2022	2	diocesan	South	5	4	4	3	4	2	5	1	1	1	3	0	2	early_compliant_survivor
54	24	2022	2	diocesan	Northeast	5	3	3	4	5	0	4	1	1	1		1		early_compliant_survivor
55	29	2017	7	diocesan	Northeast	4	4	3	3	5	0	4	1	1	1		1		early_compliant_survivor
56	28	2018	6	diocesan	South	3	5	4	2	2	3	4	1	1	1	4	0	4	late_integrated_attriter
57	26	2020	4	diocesan	West	4	4	3	4	4	0	4	1	1	1		1		early_compliant_survivor
58	23	2023	1	diocesan	West	5	3	4	4	4	2	5	1	0	1	2	0	2	early_compliant_survivor
59	26	2020	4	diocesan	South	2	1	3	2	2	1	3	0	0	0	2	0	1	early_exit_discerner
60	24	2022	2	diocesan	Midwest	5	4	4	3	5	2	5	1	0	1	2	0	2	early_compliant_survivor
61	31	2015	9	religious	South	4	2	1	4	5	1	3	1	1	0		1		institutional_careerist
62	23	2023	1	diocesan	Midwest	4	4	4	4	5	2	4	1	0	1	2	0	2	early_compliant_survivor
63	26	2020	4	diocesan	West	5	3	3	3	5	1	4	1	1	1		1		early_compliant_survivor
64	26	2020	4	diocesan	Midwest	5	3	3	4	5	0	4	1	1	1		1		early_compliant_survivor
65	27	2019	5	diocesan	West	5	4	3	3	4	1	4	1	1	1		1		early_compliant_survivor

Dashboard Structure and Visualizations

The Power BI environment consisted of a single dashboard reporting page organized around institutional trajectory and outcome analysis.

Dashboard reporting views included:

- ordination outcomes by formation trajectory
- exit reason distribution across trajectory typologies
- average compliance score by profile cluster
- leadership assignment rates by profile cluster

- year-2 advancement rates by profile cluster
- average introspection level by ordination outcome

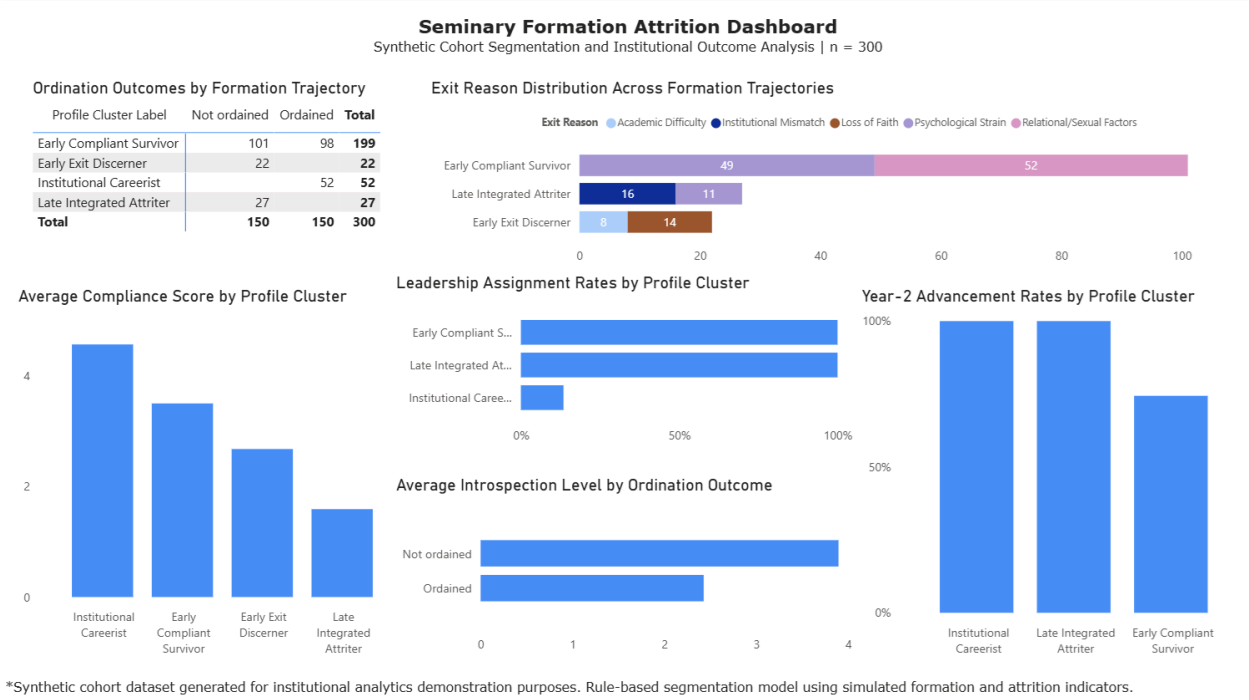
The dashboard environment used:

- grouped variable families
- manually formatted percentage displays
- DAX-based label formatting
- categorical comparison visuals
- profile-cluster segmentation structures

Dashboard visuals were configured for briefing-oriented institutional reporting rather than exploratory statistical analysis.

Figure 3 documents the Power BI dashboard environment and reporting layout.

Figure 3) Power BI Dashboard Overview



*Synthetic cohort dataset generated for institutional analytics demonstration purposes. Rule-based segmentation model using simulated formation and attrition indicators.

Validation Procedures

Validation procedures focused on internal consistency and reporting stability rather than statistical inference.

Review procedures included:

- contradictory outcome checks
- invalid exit-condition checks

- advancement consistency review
- trajectory typology distribution review
- dashboard segmentation review
- pivot-table validation checks

Specific checks ensured:

- ordained cases did not contain exit records
- non-ordained cases populated attrition variables
- early-exit trajectories did not generate advanced institutional outcomes
- dashboard visualizations remained internally consistent during filtering and segmentation review
- Pivot tables were used as an intermediate validation layer prior to dashboard import and visualization.
- Generator logic was revised iteratively during development to improve alignment between synthetic outputs and published seminary reporting patterns.

Power BI Configuration Notes

The final static dataset was imported directly into Power BI for dashboard reporting and visualization.

Power BI configuration procedures included:

- field grouping by variable category
- manual label renaming
- DAX-based display labels
- categorical formatting adjustments
- percentage formatting for advancement indicators
- dashboard visual alignment and layout adjustments

Known Constraints in Synthetic Logic

Several implementation constraints affect interpretation of the synthetic environment.

- trajectory typology weighting affects downstream outcome distributions
- rule-based generation logic reduces stochastic variability
- bounded scoring ranges constrain edge-case generation
- advancement logic is partially outcome-linked by design
- dashboard outputs prioritize segmentation readability over statistical realism
- behavioral indicators function as synthetic analytical proxies rather than validated psychometric measures

The dataset was developed for institutional trajectory analysis and dashboard-reporting demonstration purposes within a synthetic analytical environment.